

Lab 9: Conditions, Loops, next and break

Level: Intermediate Mapping: LO5 | K6 | A7 | A9 Concepts: if/else, for, while, next, break, logical operators

Scenario

A quality team scans daily temperatures and handles invalid records.

Goal

A controlled processing loop with an exception log.

Files in this folder

- lab-09.R - completed guided R script
- verify.R - repeatable acceptance check
- lab-workbook.xlsx - brief, input data, controls and evidence record
- input-data.csv - portable fallback if readxl is unavailable

Detailed step-by-step

1. Import daily readings from Excel.
2. Use a for loop to count valid days above the threshold.
3. Use next for missing readings and record them.
4. Use break only when the safety-stop value appears.
5. Compare the loop result with a vectorised validation check.

Run and verify

1. Set the RStudio working directory to this lab folder.
2. Open lab-09.R and read the header, inputs and validation rules.
3. Run the complete script with `source("lab-09.R")`.
4. Confirm the Console ends with LAB VERIFIED: outputs created and checks passed.
5. Run `source("verify.R")` and retain the verification message.
6. Open analysis_output.csv and compare the evidence with the workbook Control sheet.
7. Complete the workbook Evidence sheet and save it without changing the filename.

Acceptance criteria

- [] Missing values are skipped and counted.
- [] Safety stop is explicit and testable.
- [] Loop and vectorised count agree before the stop point.

Troubleshooting

- there is no package called 'readxl': the script automatically uses input-data.csv; install readxl later if Excel import is required.

- cannot open file: confirm RStudio's working directory is this exact lab folder.
- object not found: run the script from the top so imports and validations occur before analysis.
- Verification fails: inspect the first error, compare column names with Input_Data, and rerun from a clean session.

Evidence to submit

Completed Excel Evidence sheet, R script, generated output files and the final Console verification message.